

Career Summary

Backend engineer with 2+ years at Zoho Corporation building high-availability distributed systems in Java. Currently completing an MSc in Computer Science (Future Networked Systems) at Trinity College Dublin, with hands-on experience across microservices, cloud-native infrastructure, and full-stack development.

Experience

Member Technical Staff – Zoho Corporation, Chennai, India

06/2023 – 08/2025

- Architected and engineered an internal **Java-based platform framework** (Mickey) used by multiple product teams, reducing development time and improving long-term maintainability.
- Enhanced **failover and recovery mechanisms** by introducing automated upgrade workflows and improving server-to-server file transfer logic, reducing manual intervention during failovers by **~60%** and recovery time by **~35%**.
- Designed and developed a **four-node High-Availability and Disaster Recovery topology** using configuration-driven scheduling and Robocopy-based replication, achieving **99.95% uptime** in distributed production environments.
- Built and owned **FileSync**, a standalone **Java NIO-based** file replication utility with robust error handling, reducing file transfer error rates from **22% to 7%**, and integrated it into the HA module.
- Contributed to the design and tuning of **distributed system architectures**, supporting deployments across **100+ nodes** through scalability planning and performance optimization.
- Integrated **ClickHouse** as an optional analytics-optimized database backend into the Mickey framework, enabling teams with analytics-heavy workloads to leverage ClickHouse's fast columnar reads as an alternative to the default PostgreSQL/MSSQL storage layer.
- Scaled the Mickey framework to **20+ internal teams**, directly contributing to a **~10% increase in customer growth** across adopting teams through improved platform reliability and reduced onboarding friction.
- Mentored two summer interns on **high availability and distributed systems architecture** – guiding them through HA topology design, fault-tolerance patterns, and production deployment practices.

Project Trainee – Zoho Corporation, Chennai, India

01/2023 – 05/2023

- Architected and delivered a **test automation framework** for Java Swing applications, reducing manual testing effort by **60%** and expanding regression coverage by **40%**.
- Led the development of a **Java NIO filesystem** provider, simplifying **HDFS** integration for 15+ applications and enhancing team efficiency by **20%** through unified data access.

Summer Intern – Zoho Corporation, Chennai, India

04/2022 – 09/2022

- Built and optimized a **SQLite-backed queue mechanism** in Java to manage tasks and messages in FIFO order.
- Investigated and resolved **500+ production and support tickets** over a 6-month internship, contributing to improved system stability and reduction of recurring defects.

Projects

SmartEnoughCity – Cloud-Native Urban Transport Intelligence Platform (*Team Project, CS7CS3, TCD*)

- **Tech Stack:** Java 17, Spring Boot 3.5, Python 3.12, FastAPI, scikit-learn, React 19, TypeScript, PostgreSQL 18 + PostGIS, Docker, Kubernetes (GKE), Helm, Terraform, Istio, GitHub Actions, Prometheus, Grafana, Loki, Open-Telemetry
- Built a **4-service microservices platform** ingesting real-time data from 15+ Dublin transport APIs (GTFS, GTFS-RT, GBFS, Irish Rail XML, Luas XML) into PostgreSQL/PostGIS, serving ML-driven mobility indicators, disruption alerts, and city-planning tools through an interactive React dashboard.
- Implemented **scikit-learn ML models** (logistic regression for Dublin Bikes empty/full risk, linear regression for train ridership forecasting) retrained daily and scored every 5 minutes via APScheduler, with results served through a FastAPI inference engine.
- Provisioned the full GKE deployment with **Terraform** (6 staged modules: GKE cluster, Istio service mesh, cert-manager, ingress-nginx, network policies) and packaged all services as a **Helm umbrella chart**; CI/CD via GitHub Actions with path-filtered builds, GHCR image publishing, and GCP Workload Identity Federation (no long-lived credentials).
- Maintained a **~600-test suite** (293 pytest, 262 JUnit with Testcontainers) with a 75% coverage gate enforced on every PR, developed under an XP/Agile process with TDD, Spotless, ErrorProne, and ruff.

Distributed Traffic Booking System – CS7NS6, Trinity College Dublin

- **Tech Stack:** Python 3.12, FastAPI, SQLAlchemy 2.0 (async), PostgreSQL 15, Redis 7, RabbitMQ 3.12, React 18, TypeScript, Docker, AWS EC2/RDS, pytest
- Architected a **6-service microservices platform** with a Saga orchestrator for cross-region journey bookings across three geographic data partitions (Ireland, UK, France), coordinating reservations via peer HTTP calls and issuing compensating transactions on any failure.

- Implemented **Redis distributed locks** (SET NX PX) per road segment + time slot to prevent double-booking under concurrent load, with REPEATABLE READ isolation on all DB sessions to prevent read skew during saga capacity checks.
- Built a **Dijkstra's shortest-path routing engine** over a seeded road-segment graph with Haversine edge weights, automatically triggering the multi-region saga path for cross-border segments.
- Engineered a **saga reconciler** background job that auto-heals three classes of partition inconsistency (orphaned sagas, stuck in-progress bookings, lingering reservations) on startup and every 120 seconds; deployed on a multi-VM AWS topology with event-driven Notification and Analytics consumers via RabbitMQ.

DevVault – Full-Stack AI-Powered Developer Bookmark Manager

- **Tech Stack:** Java 17, Spring Boot 3.2, Spring Security, PostgreSQL 16, Google Gemini 2.5 Flash, React 18, TypeScript, TanStack Query, Docker, Kubernetes (GKE), GitHub Actions CI/CD
- Built a production-deployed bookmark manager with AI-powered auto-tagging and summarization: on every save, the app scrapes the page via Jina AI Reader and asynchronously calls Google Gemini 2.5 Flash to generate paragraph summaries and 3–5 tags without blocking the HTTP response.
- Engineered an event-driven AI pipeline using Spring `@TransactionalEventListener` and a dedicated thread pool, ensuring AI latency is fully decoupled from user-facing response time; a startup reconciler reprocesses any bookmarks stuck in PENDING/FAILED states.
- Deployed end-to-end on GKE with multi-stage Docker builds (JDK→JRE, Node→nginx), Flyway schema migrations, GitHub Actions CI/CD (SpotBugs static analysis + Vitest), and Kubernetes Secrets for credentials management.
- Implemented stateless JWT auth (15-min access token + 7-day refresh) alongside Google OAuth2 SSO, a friend/share social layer with a notification bell, and a public Explore feed – all documented via SpringDoc OpenAPI.

EliteCode – Adaptive Coding Interview Preparation Platform

- **Tech Stack:** Python 3.10, FastAPI, Supabase (PostgreSQL + Auth), Judge0 CE (Docker), React 18, TypeScript, Vite, Monaco Editor, scikit-learn (BKT), GitHub Actions, GCP (e2-small VM), Cloudflare Pages
- Designed and implemented a **Bayesian Knowledge Tracing (BKT)** engine in FastAPI that continuously models each user's mastery per topic and difficulty, recommending problems in their Zone of Proximal Development (50–80% predicted success) – an algorithm typically found in academic EdTech research, applied to a production system.
- Built stereotype-based onboarding priors (CS degree, bootcamp, self-taught, career-switch) with per-topic deltas and a time-on-task slip penalty, producing accurate mastery estimates from the very first session.
- Self-hosted **Judge0 CE** in Docker on a GCP VM (bound to localhost), enabling sandboxed code execution across Python, JavaScript, TypeScript, Java, C++, and Go with zero per-execution cost; test cases submitted in parallel via `asyncio.gather`.
- Implemented JWKS-based JWT verification (ES256) against Supabase Auth with Google/GitHub OAuth, Row-Level Security on all tables, and a full CI/CD pipeline (GitHub Actions SSH deploy + systemd restart / Cloudflare Pages wrangler build).

Skills

Programming Languages: Java, Python, TypeScript, C

Databases: PostgreSQL, MySQL, MSSQL, MongoDB, ClickHouse, SQLite

Backend & Web: Spring Boot, FastAPI, Node.js, Flask, React, REST APIs

Distributed Systems: High Availability, Fault Tolerance, Replication, Saga Pattern, Event-driven Architecture

Cloud & DevOps: Docker, Kubernetes, GCP GKE, AWS EC2/RDS, Helm, Terraform, GitHub Actions CI/CD

Networking: TCP/IP, DNS, HTTP

Tools: Git, GitHub, Gradle, Maven, Ant

Core CS: Data Structures & Algorithms, OOP, Operating Systems, Linux

Education

Trinity College Dublin

09/2025 – Present

Master's in Computer Science – Future Networked Systems

Grade 1:1 Expected

St. Joseph's College of Engineering, India

06/2019 – 04/2023

Bachelor's in Computer Science

Final grade 1:1 (92.3%)

Certifications & Achievements

- Awarded **Mickey Emerging Programmer (Q4 FY23)** for high-impact contributions to the internal framework.
- Recipient of **LRG Naidu Educational Trust Scholarship (ELGI Scholar)** for undergraduate academic excellence.
- **AI Infrastructure and Operations Fundamentals** – NVIDIA May 2026
- **Architecting Scalable Cloud AI Infrastructure** – Coursera Jun 2026
- **Building AI Agents and Agentic Workflows Specialization** – IBM May 2026
- **Google AI Specialization** – Google May 2026